

Tutorial No: 3

Activity Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

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Program:

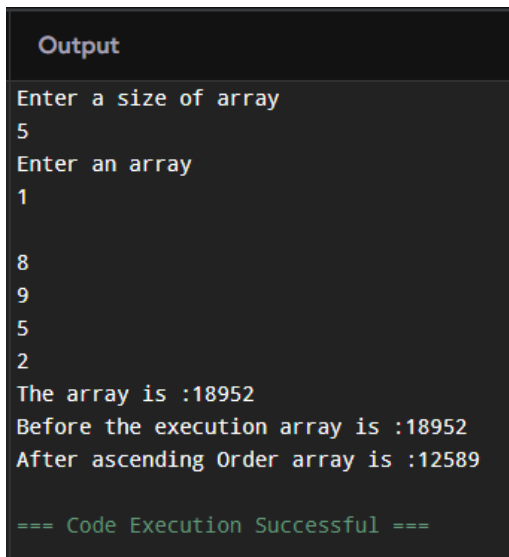
```
#include <iostream>
using namespace std;
void swap(int &a, int &b) {
    int temp=a;
    a=b;
    b=temp;
}
void ascending(int array[],int size) {
    for(int i=0;i<size-1;i++) {
        for(int j=0;j<size-1;j++) {
            if (array[j]>array[j+1]) {
                swap(array[j],array[j+1]);
            }
        }
    }
}
void descending(int array[], int size) {
    for(int i=0;i<size-1;i++) {
        for(int j=0;j<size-1;j++) {
            if (array[j]<array[j+1]) {
                swap(array[j],array[j+1]);
            }
        }
    }
}
int main() {

    int array[100],size,i;
    cout << "Enter a size of array" << endl;
    cin>>size;

    cout << "Enter an array"<<endl;
    for(i=0;i<size;i++) {
        cin>>array[i];
    }
}
```

```
cout << "The array is :";  
for(i=0;i<size;i++) {  
    cout << array[i];  
}  
cout << endl;  
cout<<"Before the execution array is :";  
for(i=0;i<size;i++) {  
    cout << array[i];  
}  
cout << endl;  
cout << "After ascending Order array is :";  
ascending(array, size);  
for(i=0;i<size;i++) {  
    cout << array[i];  
}  
return 0;  
}
```

Output:



The screenshot shows a dark-themed window titled "Output". It displays the following text: "Enter a size of array", "5", "Enter an array", "1", "8", "9", "5", "2", "The array is :18952", "Before the execution array is :18952", "After ascending Order array is :12589", and "=== Code Execution Successful ===".

```
Output  
Enter a size of array  
5  
Enter an array  
1  
8  
9  
5  
2  
The array is :18952  
Before the execution array is :18952  
After ascending Order array is :12589  
  
=== Code Execution Successful ===
```